REAPER-Notator v2.58

Comprehensive Codebase Volume & Structural Analysis

TOTAL LINES	EXECUTABLE CODE	SOURCE FILES	PATTERN LIBRARY
86,237	73,828	93 Lua	1,500 JSON

1. Architecture & Module Volume Breakdown

Module Directory	Files	Total Lines	Code Lines	Comments	Blank	Code Ratio
modules/services	38	45,461	39,226	2,686	3,746	86.3%
modules/rendering	7	15,190	13,113	878	1,248	86.3%
modules/ui	25	13,983	11,496	987	1,526	82.2%
tests	2	4,316	3,769	250	344	87.3%
modules	3	2,682	2,364	483	164	88.1%
modules/input	2	2,365	2,091	96	180	88.4%
Root	1	1,450	1,156	185	113	79.7%
modules/classes	15	790	613	110	96	77.6%
TOTAL	**93**	**86,237**	**73,828**	**5,675**	**7,417**	**85.6%**

2. Key Core System Files by Volume (Top 15)

File Name	Subsystem / Module	Lines	Code	Comments
midi_service.lua	modules/services	7,534	6,658	287
score_canvas.lua	modules/rendering	5,368	4,707	251
engraver.lua	modules/rendering	4,890	4,252	310
melody_pathfinder_service.lua	modules/services	3,182	2,715	229
test_all_modules.lua	tests	2,993	2,514	224
record_quantize_service.lua	modules/services	2,856	2,459	193
slur_service.lua	modules/services	2,826	2,481	127
canvas_context_menus.lua	modules/rendering	2,401	2,107	105
musicxml_import_service.lua	modules/services	2,193	1,911	125
key_signature_service.lua	modules/services	1,735	1,433	173
musicxml_export_service.lua	modules/services	1,715	1,452	145
portamento_service.lua	modules/services	1,683	1,440	93
glissando_service.lua	modules/services	1,594	1,360	82
mouse_handler.lua	modules/input	1,554	1,386	54
audio_preview.lua	modules/services	1,549	1,269	166

3. Curated Orchestral Pattern Library (1,500 Patterns)

Category Folder	Genre / Instrumentation Scope	Pattern Count
01_Strings_Staccato	Cinematic Strings (Staccato / Spiccato Ostinatos)	150 patterns
02_Strings_Pizzicato	Orchestral Strings (Pizzicato / Bartók Snap)	150 patterns
03_Brass_Blockbuster	Epic Brass (Horns, Trumpets, Trombones, Tubas)	150 patterns
04_Cinematic_Melodies	Heroic Themes & Symphonic Lead Melodies	150 patterns
05_Counter_Melodies	Polyphonic Counter-Melodies & Descant Voices	150 patterns
06_Woodwinds_Textures	Woodwind Runs, Flutter-Tongue & Textures	150 patterns
07_Cinematic_Piano	Concert Grand Piano (Arpeggios & Ostinatos)	150 patterns
08_Ancient_Harp_Greek_Roman	Antiquity Concert Harp (Greek & Roman Hymns)	150 patterns
09_Orchestral_Percussion	Orchestral & Cinematic Percussion (Timpani, Taiko, Drums)	150 patterns
10_Low_Strings_Celli_Basses	Low Strings Tension & Rhythm (Staccato Celli & Double Basses)	150 patterns
TOTAL PATTERNS	**10 Curated Production Categories**	**1,500 patterns**

4. Specialized Orchestral & Engraving Services Subsystem

Service Subsystem File	Musical & Architectural Responsibility	Code Lines
annotation_service.lua	Annotation Layer Engine - Per-track multi-layer free	972
circle_of_fifths_service.lua	Circle of Fifths & Harmonic Palette - Chord wheel, t	680
melody_pathfinder_service.lua	Melodic Pathfinder Engine - Corpus-trained Viterbi i	2,715
melody_style_corpus.lua	Composer Style Corpus - Statistical Markov transitio	472
divisi_service.lua	Divisi Engine - Split polyphonic voices onto full du	434
divisio_vocum_service.lua	Divisio Vocum - Orchestral Voice Division across tra	308
dynamics_engine.lua	Orchestral Dynamics Engine - Bezier/Linear CC automa	1,128
filter_notes_service.lua	Filter Irregular Notes - Metric onset analysis and v	308
frame_timer_service.lua	Native High-Refresh Timer - 1ms kernel multimedia ti	147
glissando_service.lua	Glissando Engine - Smooth chromatic / diatonic slide	1,360
imgui_guard_service.lua	ImGui Stack Guard Engine - Zero-leak Begin/End & Beg	358
key_signature_service.lua	Key Signature Service - Project/Take key signatures	1,433
midi_service.lua	Core MIDI Engine - Low-latency take I/O and event sy	6,658
musicxml_export_service.lua	MusicXML 4.0 Exporter - Standard score interchange f	1,452
musicxml_import_service.lua	MusicXML 4.0 Importer - Orchestral part loading and	1,911
audio_preview.lua	Audition Preview Engine - Non-destructive note click	1,269
reaticulate_parser.lua	Reaticulate Bridge - FX Slot 1 setup, bank sync & ke	936
record_quantize_service.lua	Record & Auto-Quantize Engine - Live MIDI capture, S	2,459
scale_service.lua	Scale & Harmony Service - Chord Track harmonic trans	932
slur_service.lua	Slur Engine - Ergonomic Bezier phrasing curves and k	2,481
portamento_service.lua	Portamento Service - High-resolution micro-tonal pit	1,440
pattern_service.lua	Pattern Engine - Orchestral motif browser and timeli	635
rehearsal_mark_service.lua	Rehearsal Marks - Custom/auto structural navigation	382
fermata_service.lua	Fermata Service - Score-wide pause holds and tempo m	254
pedal_service.lua	Pedal Service - Classical damper and sustain pedal C	637

Quality Assurance & Standards Status: 100% Certified

All 93 Lua files, 159 unit tests, and 1,500 pattern JSON files verified with 0 errors. Fully compliant with Gardner Read & Elaine Gould engraving standards.